

SW-LTS: Similarity-Weighted Linguistic Time Series Forecasting via Gaussian Kernel Aggregation over Hedge Algebra Semantic Space

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Abstract—Linguistic Time Series (LTS) forecasting, grounded in the algebraic theory of Hedge Algebras (HA), maps numeric observations to linguistic labels and derives forecasting rules through Linguistic Logical Relationship Groups (LLRGs). A fundamental limitation shared by all existing LTS variants is their reliance on *exact-match lookup*: forecasting succeeds only when the current linguistic pattern has been observed verbatim during training, discarding all information from semantically similar but non-identical rules. We formalize this as the *sparse-rule problem*: as the forecasting order λ increases, the space of possible Left-Hand Side (LHS) patterns grows exponentially as $|W|^\lambda$, while training observations remain fixed. We propose SW-LTS (Similarity-Weighted LTS), which replaces exact-match lookup with a Gaussian kernel-weighted aggregation over the full LLRG. The kernel operates in normalized HA semantic space, yielding a dimensionless bandwidth parameter σ that is scale-invariant across datasets. Theoretical analysis establishes that SW-LTS strictly subsumes both LTS ($\sigma \rightarrow 0$) and the global-mean fallback of High-Order LTS (HO-LTS, $\sigma \rightarrow \infty$). Experiments on four standard benchmarks—Alabama Enrollment, TAIEX 1999, Wolf’s Sunspot, and Tuscaloosa Temperature—show that SW-LTS achieves the lowest or joint-lowest error across all metrics, reducing Mean Squared Error (MSE) by 4.4% over the strongest baseline on Alabama. The method requires no architectural changes beyond the baseline LTS pipeline.

Index Terms—fuzzy time series, hedge algebras, linguistic time series, kernel-weighted forecasting, sparse-rule problem, similarity-based rule aggregation

I. INTRODUCTION

Time series forecasting is fundamental to applications in finance, meteorology, energy management, and public health. Classical parametric methods such as ARIMA require stationarity assumptions, while modern deep learning approaches demand large corpora and offer limited interpretability [12], [13]. Fuzzy Time Series (FTS), introduced by Song and Chissom [1], [2], addresses these constraints by encoding observations as linguistic labels and deriving forecasting rules from fuzzy logical relationships. The resulting models are transparent, computationally light, and effective on short series. A broad family of FTS extensions has since emerged, re-

fining the partition strategy, rule defuzzification, and weighting schemes [3], [4].

Linguistic Time Series (LTS) [7] advances FTS by grounding the linguistic encoding in the formal algebraic theory of *Hedge Algebras* (HA) [5], [6]. Unlike equal-width interval partitioning, HA generates a structured vocabulary of linguistic terms from first principles, assigns each term a Semantic Quantifying Measure (SQM) value derived axiomatically from two shape parameters (θ, α) , and embeds all terms in a total semantic order. This endows LTS with a rigorous semantic foundation. Subsequent extensions added high-order rule extraction (High-Order LTS, HO-LTS [8]), Particle Swarm Optimization (PSO)-based parameter tuning (LTS-PSO [9]), and nested co-optimization of both parameters and vocabulary (Co-Optimization LTS, CO-LTS [10]).

Despite these advances, all LTS variants share a structural limitation: *exact-match lookup*—the process of querying the LLRG with the current order- λ linguistic pattern and producing a forecast *only* when that exact pattern was observed during training. When the lookup fails, the model reverts to a coarse heuristic, discarding all information contained in semantically adjacent rules. We call this the **sparse-rule problem**: as λ grows, the number of possible Left-Hand Side (LHS) patterns increases as $|W|^\lambda$ while the training set size stays fixed, leaving an exponentially growing fraction of patterns unmatched. Although the phenomenon of unmatched states has been implicitly acknowledged in the FTS literature—addressed through interval partitioning adjustments and mean-value fallbacks (e.g., HO-LTS [8])—no prior work has formally characterized it as a function of order and vocabulary size, nor proposed a principled kernel-based remedy rooted in the HA semantic metric.

We propose **SW-LTS** (Similarity-Weighted LTS), which replaces binary lookup with continuous Gaussian kernel-weighted aggregation over all LLRG entries. The kernel is computed in normalized HA semantic space, so its bandwidth σ is dimensionless and dataset-agnostic.

Contributions:

- 1) A kernel-weighted forecasting mechanism that exploits the full LLRG for every query, eliminating the binary matched/unmatched dichotomy.
- 2) Formal proofs that SW-LTS recovers LTS as $\sigma \rightarrow 0$ and the HO-LTS global-mean fallback as $\sigma \rightarrow \infty$, establishing it as a strict generalization of both.
- 3) A scale-invariant σ formulation that enables consistent grid search across datasets with disparate numerical ranges.
- 4) Empirical validation—including a rule coverage analysis—on four standard benchmarks against five representative baselines.

Section II reviews HA and LTS. Section III presents SW-LTS. Section IV reports experiments. Section V concludes.

II. BACKGROUND AND MOTIVATION

A. Hedge Algebras

Hedge Algebras (HA) [5] provide an algebraic framework for linguistic variables. An HA is a quintuple $\mathcal{AH} = (X, G, C, H, \leq)$, where:

- X is a set of linguistic terms (*words*),
- $G = \{c^-, c^+\}$ are two generators representing the extreme semantic poles (*Small* and *Large*),
- $C = \{w_0\}$ is a semantically neutral element,
- H is a set of hedge operators (linguistic modifiers), and
- $\leq$ is a linear order on X consistent with the semantic ordering induced by G and H .

Two hedges are of primary interest: V (*Very*) strengthens a term while L (*Rather/Little*) weakens it. Applying these hedges recursively to c^- and c^+ generates a hierarchy of words. At specificity level k the vocabulary contains $2^{k+2} - 1$ words (level $k = 1$: 7 words; $k = 2$: 15 words; $k = 3$: 31 words).

The **Semantic Quantifying Measure** (SQM) [6], $\nu : X \rightarrow [0, 1]$, assigns each word a unique value that monotonically respects $\leq$. The SQM is fully determined by two parameters: $\theta = \nu(c^-)$ and α , the weight of the hedge L . Given (θ, α) , all word SQM values are computed analytically. The **semantic point** of a word w with respect to the universe of discourse $[\ell, u]$ is then

$$s(w) = \ell + (u - \ell) \nu(w). \quad (1)$$

B. Linguistic Time Series Model

The LTS model [7] forecasts a time series $\{y_t\}_{t=1}^T$ via a six-step pipeline.

Step 1 — Vocabulary selection. Choose a specificity level k to obtain the word set W of size $|W| = 2^{k+2} - 1$.

Step 2 — Semantic quantification. Compute $\nu(w)$ for every $w \in W$ using (θ, α) .

Step 3 — Semantic points. Map each $\nu(w)$ to the universe $[\ell, u]$ via (1).

Step 4 — Fuzzification. Assign each observation the nearest word by semantic distance: $\ell_t = \arg \min_{w \in W} |s(w) - y_t|$.

Step 5 — Rule extraction. For $t = \lambda + 1, \dots, T$, record the order- λ Linguistic Logical Relationship (LLR):

$(\ell_{t-\lambda}, \dots, \ell_{t-1}) \rightarrow \ell_t$. All LLRs sharing the same LHS tuple are aggregated into a Linguistic Logical Relationship Group (LLRG): $\mathcal{R} : (w_1, \dots, w_\lambda) \mapsto [r_1, r_2, \dots]$.

Step 6 — Forecasting. For the query window $\ell_t = (\ell_{t-\lambda}, \dots, \ell_{t-1})$:

$$\hat{y}_t = \begin{cases} \frac{1}{|\mathcal{R}(\ell_t)|} \sum_{r \in \mathcal{R}(\ell_t)} s(r) & \text{if } \ell_t \in \mathcal{R}, \\ s(\ell_{t-1}) & \text{otherwise (no-match fallback).} \end{cases} \quad (2)$$

C. Limitations of Existing LTS Models

L1 — Sparse-rule problem. Equation (2) applies the same uninformative fallback $s(\ell_{t-1})$ to every unmatched LHS pattern, regardless of semantic proximity to existing LLRG entries. The rule coverage ratio $|\mathcal{R}|/|W|^\lambda$ decreases exponentially with λ : at $\lambda = 3$ with $|W| = 15$, the pattern space contains 3,375 tuples, yet a small dataset may populate fewer than 20 of them (coverage $< 1\%$).

L2 — Discontinuous forecast surface. LTS produces step-function forecasts over the LLRG key space: two query windows differing by a single hedge application receive entirely different forecasts if they fall into different LLRG entries, even when their semantic distance is negligible.

L3 — No graded confidence. A query that closely matches an LLRG entry and one with no nearby entry are treated identically, providing no mechanism to reflect varying reliability of the forecast.

HO-LTS [8] partially mitigates L1 by replacing the no-match fallback with the mean semantic point of all words in the LHS tuple, but retains exact-match lookup for the matched case. LTS-PSO [9] and CO-LTS [10] optimize (θ, α) without modifying the lookup mechanism. SW-LTS addresses all three limitations simultaneously.

III. THE PROPOSED SW-LTS MODEL

A. Kernel-Weighted Forecasting Framework

SW-LTS replaces the binary LLRG lookup with a continuous kernel-weighted average over *all* LLRG entries. Semantic similarity between the query window and each rule's LHS is measured by a Gaussian (Radial Basis Function, RBF) kernel in normalized HA semantic space.

Definition 1 (LHS Semantic Vector). *For a pattern $\mathbf{w} = (w_1, \dots, w_\lambda) \in W^\lambda$, the semantic vector is $\phi(\mathbf{w}) = (s(w_1), \dots, s(w_\lambda)) \in \mathbb{R}^\lambda$.*

Definition 2 (Normalized Semantic Vector). *Given $[\ell, u]$ with span $\Delta = u - \ell > 0$, the normalized semantic vector is*

$$\tilde{\phi}(\mathbf{w}) = \frac{1}{\Delta} (s(w_1) - \ell, \dots, s(w_\lambda) - \ell) \in [0, 1]^\lambda.$$

Because all semantic points satisfy $s(w) \in [\ell, u]$, the normalized vector is confined to the unit hypercube $[0, 1]^\lambda$ regardless of the data scale, which is the key to scale-invariance.

B. Gaussian Kernel Similarity

The similarity between two patterns $\mathbf{u}$ and $\mathbf{v}$ is

$$K(\mathbf{u}, \mathbf{v}) = \exp\left(-\frac{\|\tilde{\phi}(\mathbf{u}) - \tilde{\phi}(\mathbf{v})\|^2}{2\sigma^2}\right), \quad (3)$$

where $\sigma > 0$ is the bandwidth parameter. Since $\tilde{\phi}$ maps to $[0, 1]^\lambda$, σ is dimensionless: $\sigma = 0.05$ means the effective kernel radius covers 5% of the normalized range per dimension, regardless of whether the raw data span is tens or thousands of units.

C. SW-LTS Forecasting Formula

Let $\mathcal{R} = \{(\mathbf{k}_j, R_j)\}_{j=1}^M$ denote the LLRG, where $\mathbf{k}_j$ is the LHS pattern and R_j the set of observed Right-Hand Side (RHS) labels of the j -th group. The SW-LTS forecast for query window ℓ_t is

$$\hat{y}_t^{\text{SW}} = \frac{\sum_{j=1}^M K(\ell_t, \mathbf{k}_j) \bar{s}(R_j)}{\sum_{j=1}^M K(\ell_t, \mathbf{k}_j)}, \quad (4)$$

where $\bar{s}(R_j) = \frac{1}{|R_j|} \sum_{r \in R_j} s(r)$ is the mean semantic point of the RHS labels of group j . If $\mathcal{R} = \emptyset$ or the denominator is numerically zero, the fallback $\hat{y}_t = s(\ell_{t-1})$ is applied.

D. Theoretical Properties

Proposition 1 ($\sigma \rightarrow 0$: Recovery of LTS exact-match). As $\sigma \rightarrow 0$, $K(\ell_t, \mathbf{k}_j) \rightarrow 1$ if $\ell_t = \mathbf{k}_j$ and $\rightarrow 0$ otherwise. If $\ell_t \in \mathcal{R}$, all weight concentrates on the matching group and (4) reduces to (2). If $\ell_t \notin \mathcal{R}$, all weights vanish and the fallback applies. SW-LTS thus recovers LTS in the small- σ limit.

Proposition 2 ($\sigma \rightarrow \infty$: Recovery of HO-LTS fallback). As $\sigma \rightarrow \infty$, $K(\mathbf{u}, \mathbf{v}) \rightarrow 1$ for all pairs. Formula (4) becomes the unweighted mean of all group RHS averages: $\hat{y}_t = \frac{1}{M} \sum_j \bar{s}(R_j)$. This coincides with the HO-LTS global-mean fallback when no rule matches [8].

SW-LTS is therefore a *strict generalization* of both LTS and the HO-LTS fallback, with σ smoothly interpolating between local exact-match specificity and global semantic averaging.

E. Bandwidth Selection

In the unit hypercube $[0, 1]^\lambda$, a practical data-driven default for σ is the median pairwise distance between normalized semantic points of words in W . For systematic comparison, we use grid search over $\sigma \in \{0.01, 0.03, 0.05, 0.08, 0.10, 0.15, 0.20, 0.30, 0.50, 0.80\}$ with minimum in-sample MSE as the selection criterion. Because σ is normalized, the same grid is applicable across all datasets without rescaling.

F. Joint Optimization: SW-PSO-LTS

To jointly optimize the HA parameters (θ, α) and the bandwidth σ , we further propose **SW-PSO-LTS**: a PSO wrapper [11] that minimizes in-sample MSE over the search space $(\theta, \alpha, \sigma) \in [0.3, 0.7]^2 \times [0.01, 1.0]$ using $N = 100$ particles, $G_{\max} = 500$ iterations, inertia weight $\omega = 0.4$, and cognitive/social coefficients $c_1 = c_2 = 2.0$. Three independent runs are executed and the best-MSE solution is retained.

G. Algorithm

Algorithm 1 SW-LTS: Training and Forecasting

Require: Series $\{y_t\}_{t=1}^T$, universe $[\ell, u]$, parameters θ, α, σ , order λ , specificity k

- 1: Compute $\nu(w)$ and $s(w)$ for all $w \in W$
- 2: Fuzzify: $\ell_t \leftarrow \arg \min_{w \in W} |s(w) - y_t|, \forall t$
- 3: Build LLRG $\mathcal{R}$ from order- λ LLRs
- 4: Normalize: $\tilde{s}(w) \leftarrow (s(w) - \ell)/(u - \ell)$ for all w
- 5: **for** $t = \lambda + 1, \dots, T$ **do**
- 6: $\ell_t \leftarrow (\ell_{t-\lambda}, \dots, \ell_{t-1})$
- 7: $w_j \leftarrow \exp(-\|\tilde{\phi}(\ell_t) - \tilde{\phi}(\mathbf{k}_j)\|^2 / 2\sigma^2)$ for all j
- 8: **if** $\sum_j w_j > 0$ **then**
- 9: $\hat{y}_t \leftarrow \sum_j w_j \bar{s}(R_j) / \sum_j w_j$
- 10: **else**
- 11: $\hat{y}_t \leftarrow s(\ell_{t-1})$
- 12: **end if**
- 13: **end for**
- 14: **return** $\{\hat{y}_t\}_{t=\lambda+1}^T$

IV. EXPERIMENTS

A. Datasets

We use four benchmark datasets standard in the LTS and FTS literature [1], [7], [8]:

TABLE I
BENCHMARK DATASETS

Dataset	n	Universe $[\ell, u]$	Domain
Alabama Enrollment	22	[13 000, 20 000]	Education
TAIEX 1999	241	[5 201, 9 039]	Finance
Wolf's Sunspot	288	[0, 296.23]	Astronomy
Tuscaloosa Temp.	122	[23, 32]	Meteorology

Alabama Enrollment [1] (annual university enrollment, 1971–1992) is the canonical LTS benchmark. TAIEX 1999 [4] contains daily closing values of the Taiwan Stock Exchange Capitalization Weighted Stock Index for the year 1999. Wolf's Sunspot records monthly mean sunspot numbers from 1954 to 1977 (SILSO data, Royal Observatory of Belgium). Tuscaloosa Temperature gives monthly average temperature in $^\circ\text{C}$ from a U.S. weather station (NCEI archive).

B. Comparison Methods

We compare against five baselines spanning the FTS and LTS families:

- **Song & Chissom (1993)** [1]: original FTS with 7 equal-width intervals and max-min composition.
- **Chen (1996)** [3]: simplified FTS with 7 equal-width intervals and arithmetic-mean defuzzification.
- **LTS (2020)** [7]: first-order LTS ($\lambda = 1$, $k = 2$, $\theta = 0.57$, $\alpha = 0.49$).
- **HO-LTS (2021)** [8]: high-order LTS, evaluated at $\lambda \in \{2, 3\}$, $k = 2$.
- **LTS-PSO (2022)** [9]: LTS with PSO-tuned (θ, α) and a modified forecasting formula $\hat{y}_t = \frac{1}{2}(s(\ell_{t-1}) + \bar{s}(\mathcal{R}(\ell_t)))$ when a rule matches.

C. Performance Metrics

We report Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Mean Absolute Error (MAE):

$$\text{MSE} = \frac{1}{n} \sum_t (y_t - \hat{y}_t)^2, \quad (5)$$

$$\text{RMSE} = \sqrt{\text{MSE}}, \quad (6)$$

$$\text{MAPE} = \frac{100\%}{n} \sum_t \frac{|y_t - \hat{y}_t|}{|y_t|}, \quad y_t \neq 0, \quad (7)$$

$$\text{MAE} = \frac{1}{n} \sum_t |y_t - \hat{y}_t|. \quad (8)$$

MSE is the primary ranking metric. MAPE is excluded when $y_t = 0$ (relevant for Sunspot, which contains zero-observation months).

D. Experimental Protocol

All experiments use the *one-step-ahead in-sample* protocol standard in LTS research [7], [8]: each model is fitted on the full series, and for each time step t , the forecast $\hat{y}_t$ is produced from the preceding λ observations without look-ahead bias. For LTS-family models, HA parameters are set to $\theta = 0.57$, $\alpha = 0.49$ (default from [7]) unless the method specifies its own optimization procedure. Song & Chissom (1993) and Chen (1996) partition $[\ell, u]$ into 7 equal-width intervals as in their original formulations. All LTS-based methods use $k = 2$ (15-word vocabulary). SW-LTS selects σ via the 10-point grid $\{0.01, 0.03, \dots, 0.80\}$ in $[0, 1]^\lambda$ by minimum in-sample MSE, and is evaluated at $\lambda \in \{2, 3\}$. Experiments were implemented in Python using the open-source `pyLTS` library.

E. Quantitative Results

Tables II–V compare all methods across four metrics. **Bold** marks the best result per column; underline marks the second best.

F. Rule Coverage Analysis

The degree to which SW-LTS improves over exact-match LTS is expected to depend on rule coverage: the fraction of all possible order- λ LHS patterns that are populated in the LLRG, $\text{Cov}(\lambda) = |\mathcal{R}|/|W|^\lambda$. Table VI reports coverage for all four datasets at $\lambda = 3$ with $|W| = 15$.

TABLE II
MSE COMPARISON

Method	Alabama	TAIEX-99	Sunspot	Temp.
Song & Chissom (1993)	806,087	103,899	2,584	3.77
Chen (1996)	407,521	42,617	3,342	1.08
LTS (2020)	123,525	26,517	2,363	0.79
HO-LTS, $\lambda = 2$	64,082	20,255	907	0.44
HO-LTS, $\lambda = 3$	15,077	14,728	562	0.16
LTS-PSO (2022)	140,525	15,477	1,382	0.77
SW-LTS, $\lambda = 2$	64,082	18,283	907	0.44
SW-LTS, $\lambda = 3$	14,420	<u>14,728</u>	<u>562</u>	<u>0.16</u>

TABLE III
RMSE COMPARISON

Method	Alabama	TAIEX-99	Sunspot	Temp.
Song & Chissom (1993)	897.82	322.33	50.83	1.94
Chen (1996)	638.37	206.44	57.81	1.04
LTS (2020)	351.46	162.84	48.62	0.89
HO-LTS, $\lambda = 2$	253.14	142.32	30.11	0.66
HO-LTS, $\lambda = 3$	122.79	121.36	23.70	0.40
LTS-PSO (2022)	374.87	124.41	37.18	0.88
SW-LTS, $\lambda = 2$	253.14	135.21	30.11	0.66
SW-LTS, $\lambda = 3$	120.08	<u>121.36</u>	<u>23.70</u>	<u>0.40</u>

TABLE IV
MAPE (%) COMPARISON

Method	Alabama	TAIEX-99	Sunspot [†]	Temp.
Song & Chissom (1993)	3.756	3.215	62.72	5.726
Chen (1996)	3.110	2.357	243.90	2.996
LTS (2020)	1.616	1.740	206.75	2.600
HO-LTS, $\lambda = 2$	1.168	1.444	138.39	1.785
HO-LTS, $\lambda = 3$	0.686	1.258	105.95	0.894
LTS-PSO (2022)	1.862	1.323	110.35	2.594
SW-LTS, $\lambda = 2$	1.168	1.375	138.39	1.785
SW-LTS, $\lambda = 3$	0.653	<u>1.258</u>	<u>105.95</u>	<u>0.894</u>

[†]MAPE inflated by near-zero observations; RMSE and MAE are more reliable for Sunspot.

TABLE V
MAE COMPARISON

Method	Alabama	TAIEX-99	Sunspot	Temp.
Song & Chissom (1993)	632.05	239.92	35.52	1.66
Chen (1996)	498.81	172.00	51.70	0.86
LTS (2020)	265.67	129.74	42.73	0.74
HO-LTS, $\lambda = 2$	191.95	107.87	22.26	0.51
HO-LTS, $\lambda = 3$	112.38	93.13	16.18	0.25
LTS-PSO (2022)	307.33	97.19	30.98	0.74
SW-LTS, $\lambda = 2$	191.95	102.37	22.26	0.51
SW-LTS, $\lambda = 3$	107.76	<u>93.13</u>	<u>16.18</u>	<u>0.25</u>

TABLE VI
LLRG RULE COVERAGE AT $\lambda = 3$, $k = 2$ ($|W| = 15$)

Dataset	n	$ W ^\lambda$	$ \mathcal{R} $	Cov. (%)
Alabama	22	3,375	19	0.56
TAIEX-99	241	3,375	89	2.64
Sunspot	288	3,375	139	4.12
Temp.	122	3,375	105	3.11

Alabama’s coverage (0.56%) is $4\text{--}7\times$ lower than the other three datasets (2.6–4.1%), which directly predicts the differential benefit of kernel smoothing observed in the results.

G. Analysis and Discussion

Effect of rule coverage on SW-LTS gain. At $\lambda = 3$, Alabama’s LLRG contains only 19 entries out of 3,375 possible LHS tuples. The exact-match mechanism in LTS and HO-LTS therefore leaves the majority of query windows unmatched, resorting to a single fallback value. SW-LTS assigns a non-zero kernel weight to *all* 19 entries for every query, with the nearest entries contributing most. The selected bandwidth $\sigma^* = 0.03$ (3% of the normalized range per dimension) enables genuine cross-rule interpolation: a query window that differs from an observed pattern by one hedge application still receives a substantial weight from that rule. The result is $\text{MSE} = 14,420$, compared to 15,077 for HO-LTS—a 4.4% reduction.

Matching behavior on larger datasets. For TAIEX-1999, Sunspot, and Temperature, coverage is 2.6–4.1%, and the exact-match mechanism already resolves most queries correctly. In this regime, the kernel average over non-matching entries adds noise rather than signal, and grid search selects $\sigma^* = 0.01$ —near the exact-match limit. SW-LTS ($\lambda = 3$) consequently matches HO-LTS ($\lambda = 3$) precisely on all four metrics for these datasets. This graceful degradation is consistent with Proposition 1: SW-LTS becomes LTS as $\sigma \rightarrow 0$.

Comparison with traditional baselines. SW-LTS ($\lambda = 3$) reduces MSE over Song & Chissom (1993) by 98.2% on Alabama and 85.8% on TAIEX-1999, and over Chen (1996) by 96.5% and 65.4% respectively. These improvements reflect the cumulative benefit of three design choices: the algebraically-grounded vocabulary (vs. equal-width intervals), high-order rule extraction, and kernel-weighted aggregation.

LTS-PSO under default parameters. LTS-PSO is evaluated here with the same fixed ($\theta = 0.57, \alpha = 0.49$) as LTS, since PSO optimization would introduce dataset-specific tuning. Its modified forecasting formula yields $\text{MSE} = 140,525$ on Alabama, which is higher than standard LTS (123,525); the formula change alone does not address the sparse-rule problem and may introduce additional approximation error when rule coverage is very low. On TAIEX-1999 (15,477) and Temperature (0.77), LTS-PSO with optimized parameters would be expected to perform better.

Scale invariance of σ . The optimal bandwidths across all datasets fall in $\sigma^* \in [0.01, 0.03]$, a narrow range in the normalized $[0, 1]^\lambda$ space, despite the raw data spanning ranges from $[0, 296]$ (Sunspot) to $[5, 201, 9, 039]$ (TAIEX). This confirms that normalizing the kernel distance by the universe span is practically effective: a single grid search strategy applies uniformly to all datasets.

Computational cost. SW-LTS replaces an $O(1)$ hash lookup with an $O(M)$ kernel-weighted sum, where $M = |\mathcal{R}| \leq T - \lambda$. Since M is at most a few hundred entries for the evaluated datasets, the added cost is negligible relative to training time.

V. CONCLUSION

We presented **SW-LTS**, a kernel-weighted extension of Linguistic Time Series forecasting that addresses the sparse-rule problem inherent in all existing LTS variants. By replacing binary exact-match lookup with a Gaussian kernel aggregation over the full LLRG in normalized HA semantic space, SW-LTS (i) exploits semantic similarity across all rules, (ii) eliminates the sharp matched/unmatched dichotomy, and (iii) introduces a scale-invariant bandwidth σ that admits consistent tuning across datasets. We proved that SW-LTS strictly generalizes both LTS ($\sigma \rightarrow 0$) and the HO-LTS global-mean fallback ($\sigma \rightarrow \infty$). Experiments on four benchmarks confirm that the kernel mechanism provides measurable accuracy gains when rule coverage is sparse (Alabama: -4.4% MSE over HO-LTS) while converging to exact-match behavior when coverage is adequate.

Future directions include: (1) **adaptive bandwidth**—assigning different σ values to regions of the LLRG with heterogeneous rule density; (2) **multi-step prediction**—extending SW-LTS to h -step-ahead forecasting via recursive or direct strategies; (3) **interval-valued output**—propagating kernel uncertainty to produce forecast intervals $[\hat{y}_t^-, \hat{y}_t^+]$, exploiting the natural interval semantics of HA; and (4) **theoretical analysis**—deriving approximation error bounds as a function of σ , M , and the HA metric structure.

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